# MILITARY SPECIFICATION
# CARTRIDGE, DELAY: M114 ASSEMBLY

## 1. SCOPE

1.1 This specification covers Cartridge, Delay: M114 Assembly for use with propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification For |
| MIL-E-5272 | - Environmental Testing, Aeronautical and Associated Equipment, General Specification For |
| MIL-C-60539 | - Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-413 | - Visual Inspection Guide for Rubber O-Rings |
| MS-28741 | - Hose, Assembly, Detachable End Fitting, Medium Pressure |



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2.1 (Cont'd)

DRAWINGS

Munitions Command

| | |
|---|---|
| D8596110 | - Cartridge, Delay: M114 |
| C8845190 | - Carton, Packing, Ammunition for Cartridge, Delay: M114 |
| C8845191 | - Box, Packing, Ammunition, for Cartridge, Delay: M114 |
| IEL-8596110 | - Inspection Equipment List for Cartridge, Delay: M114 |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and reference specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant, or derangement of any components.

3.4 Functional.

3.4.1 The cartridge shall not fail to fire.



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3.4.2 Ballistics.-The cartridge when tested in an M44 or M45 initiator shall produce pressures and delay times at the temperatures specified in Table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, size No. 4, in accordance with MS-28741 and a pressure chamber with a volume of 1 cubic inch.

TABLE I

| Temperature Degrees ( °) Fahrenheit (F) | Pressure Pounds Per Square Inch (PSI) Min. | Max. | Delay Time (Seconds) (See 6.2) |
|---|---|---|---|
| -65° to -70°F | 1300 | 2200 | 4.2 max. |
| 65° to 75°F | 1300 | 2200 | 3.0 ± .3 |
| 160° to 165°F | 1300 | 2200 | 2.3 min. |

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following.

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, coldshuts, inclusions, porosity or other metal defects.

3.5.3 Burr.-The metal part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The metal part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection .-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspectins set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.



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4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 uncoated cartridge cases for hardness tests, 23 coated cases for protective finish tests, 5 cartridges for sealant tests, 30 cartridges to be tested functionally and 10 sets of unassembled components (including explosive components) for visual and dimensional examination and shall be tested as indicated in Table II herein.

TABLE II

| Paragraphs Requirements | Test | Test | Sample Size | Type of Test Destructive | Non-Destructive |
|---|---|---|---|---|---|
| Dwg. | 4.4.1 | Hardness | 50 (uncoated) | X | - |
| 3.2 | 4.4.2 | Prot. Finish | 23 each | X | - |
| 3.3 | 4.4.3 | Vibration | 12 (from ballistics) | - | X |
| Dwg. | 4.4.4 | Sealant App. | 5 each | X | - |
| Dwg. | 4.4.5 | Airtightness | Each | - | X |
| 3.4 | 4.4.6 | Ballistics | 30 each | X | - |

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE III

| Temperature | Not Vibrated | Vibrated |
|---|---|---|
| -65° to -70° F | 6 | 4 |
| 65° to 75° F | 6 | 4 |
| 160° to 165° F | 6 | 4 |



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4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-Unless otherwise specified by the procuring agency, an ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process
b. Delay elements from not more than one lot
c. Primers from not more than one lot.
d. Propellant from not more than one lot.

e. Igniter from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. Four power (4X) magneification may be used to perform visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination.

TABLE IV

| Lot Size | Sample Size | Major | | Minor | |
|---|---|---|---|---|---|
| | | Accept | Reject | Accept | Reject |
| 1 - 300 | 100% Examination | | | | |
| 301 - 1300 | 110 | 1 | 2 | 2 | 3 |
| 1301 - 3200 | 150 | 2 | 3 | 3 | 4 |
| 3201 - 8000 | 225 | 3 | 4 | 4 | 5 |
| 8001 - 22000 | 300 | 4 | 5 | 5 | 6 |



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4.3.2 (Cont'd)

TABLE V

| | Total Allowable Defects Per Sample Size | | | |
|---|---|---|---|---|
| Sample Size for Major Defects<br>Number of Defects in Class | 110 | 150 | 225 | 300 |
| 2 - 3 | 1 | 2 | 3 | 4 |
| 4 - 7 | 2 | 3 | 4 | 5 |
| 8 - 19 | 3 | 4 | 5 | 7 |
| 20 - 43 | 4 | 5 | 8 | 10 |
| Over 43 | 6 | 8 | 11 | 14 |
| Sample Size for Minor Defects<br>Number of Defects in Class | 110 | 150 | 225 | 300 |
| 2 - 3 | 2 | 3 | 4 | 5 |
| 4 - 7 | 3 | 4 | 5 | 7 |
| 8 - 16 | 4 | 5 | 8 | 10 |
| 17 - 48 | 6 | 8 | 11 | 14 |
| Over 48 | 8 | 11 | 17 | 20 |

4.3.2.1 Classification of defects.

4.3.2.1.1 Sealing Disc, Drawing C8594077, covering a detail of Drawing D8596110.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Metal defective | Visual |
| 2. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. Diameter of body, min. | Gage |
| 102. Thickness | Gage |
| 103. Protective finish inadequate or defective | Visual |
| 104. Burr | Visual |
| 105. Foreign matter | Visual |
| MINOR | |
| 201. Diameter of body, max. | Gage |
| 202. Varnish missing or inadequate | Visual |



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4.3.2.1.2 Case, Cartridge, Drawing C8595202 Part Number 8595393, covering a detail of Drawing D8596110.

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| 1. Metal defective | Visual |
| **MAJOR** | |
| 101. Concentricity of head with body | Gage |
| 102. Straightness of base | Gage |
| 103. Diameter of body, min. | Gage |
| 104. Diameter of head, min. | Gage |
| 105. Thickness of base | Gage |
| 106. Thickness of wall | Gage |
| 107. Location of datum diameter | Gage |
| 108. Protective finish inadequate or defective | Visual |
| 109. Burr or sharp corner | Visual |
| 110. Foreign matter | Visual |
| 111. Hardness | Gage |
| **MINOR** | |
| 201. Diameter of head, max. | Gage |
| 202. Diameter of body, max. | Gage |
| 203. Total length | Gage |
| 204. Radius or chamfer missing or incorrect | Visual |
| 205. Surface finish improper | Visual |

4.3.2.1.3 Delay Element M13 Drawing C8596113 covering a detail of Drawing D8596110.

| | |
|---|---|
| **CRITICAL** | |
| 1. Mixed delay elements | Visual |
| 2. Flash hole missing or obstructed | Visual |
| 3. Primer uncrimped or seal missing | Visual |
| 4. Marking missing, incorrect or illegible | Visual |
| 5. Delay charge not visible through flash hole | Visual |
| 6. Metal defective | Visual |



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4.3.2.1.3 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MAJOR** | |
| 101. Protective finish inadequate or defective | Visual |
| 102. Burr | Visual |
| 103. Foreign matter | Visual |
| **MINOR** | |
| None defined | |

4.3.2.1.4 Cartridge, Delay, M114, Assembly, Drawing D8596110.

| | |
|---|---|
| **CRITICAL** | |
| 1. Any part missing | X-Ray/Visual 1/ |
| 2. Improper assembly | X-ray/Visual 1/ |
| 3. Mixed ammunition | Visual |
| 4. Metal defective | Visual |
| 5. Crimp missing or incorrect | Visual |
| 6. Marking missing, incorrect or illegible | Visual |
| **MAJOR** | |
| 101. Cartridge profile failure, requiring more than 20 pounds deadweight to insert in concentricity receiver gage | Gage |
| 102. Diameter of body, max. | Gage |
| 103. Diameter of head, max. | Gage |
| 104. Length to datum diameter | Gage |
| 105. Dent, failure to chamber in concentricity receiver gage with 20 pounds deadweight pressure | Gage/Visual |
| 106. O-ring split, cracked or cut | Visual |
| 107. Protective finish inadequate or defective | Visual |
| 108. Burr | Visual |
| 109. Foreign matter | Visual |
| **MINOR** | |
| 201. Total length | Gage |
| 202. Dent, which will chamber in concentricity receiver gage with 20 pounds deadweight, maximum | Gage/Visual |
| 203. Draw scratch | Visual |
| 204. Fold, wrinkle, buckle or bulge | Visual |
| 205. Varnish inadequate or missing | Visual |
| 206. Surface finish improper | Visual |

1/ The cartridge shall be X-rayed with the sides of the case perpendicular to the rays. The X-ray negatives, properly identified, shall be forwarded to the procuring activity



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4.3.2.1.5 Carton, Packing, Ammunition for Cartridge Delay, M114, Drawing C8845190.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual/Visual |
| 202. Quantity of contents incorrect | Visual |
| 203. Separator or filler missing | Visual |
| 204. Heat sealer incomplete, loose or distorted | Visual |
| 205. Cut or damaged through impregnated box | Visual |
| 206. Marking missing, incorrect or illegible | Visual |

4.3.2.1.6 Box, Packing, Ammunition for Cartridge, Delay, M114 Drawing C8845191.

| | |
|---|---|
| CRITICAL | |
| None defined | |
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual/Visual |
| 202. Hardware missing, loose, broken or improperly engaged | Visual |
| 203. Strapping missing, loose, broken or improperly assembled | Visual |
| 204. Contents of box exposed | Visual |
| 205. Box damaged | Visual |
| 206. Marking missing, incorrect or illegible | Visual |
| 207. Number of cartons incorrect | Visual |
| 208. Car seal missing, unsealed or improperly positioned | Visual |

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.-Fifty units from each heat treated lot shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.



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4.3.3.2 Protective finish.-Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.3 Vibration.-Twelve of the assembled cartridges of the initial production lot or 12 assembled cartridges from a subsequent lot when a change is made in the lot of propellant, igniter or delay elements, shall be subjected to vibration tests. After vibration, the 12 cartridges shall be x-rayed in accordance with Note 1/ of 4.3.2.1.4 for evidence of deterioration of the primer or propellant or derangement of metal parts. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.4 Sealant application.-Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.5 Airtightness.-Each cartridge assembly shall be subjected to the specified pressure differential. Failure of any cartridge to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.6 Functional.-The following test shall be performed by the Government.

4.3.3.6.1 Ballistics.-The 30 assembled cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.4.6.1.1. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8596110.

4.4 Test methods and procedures.

4.4.1 Hardness.-A hardness testing machine shall be used to determine compliance with drawing requirements.

4.4.2 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. All anodized components shall be sectioned longitudinally to better expose the critical surface to the salt spray.

4.4.3 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-C-5272, Procedure XII.

4.4.4 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.



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4.4.5 Airtightness.-The method for determining airtightness shall be by use of equipment listed on IEL-8596110.

4.4.6 Functional.

4.4.6.1 Ballistics.-The cartridges shall be loaded into acceptable M44 or M45 initiators and temperature conditioned in accordance with 4.4.6.1.1 and Table VI. The test initiators shall be rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the sample unit shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-A simulated M44 or M45 initiator loaded with an inert M114 cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be one (1) hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

TABLE VI

| Temperature | Sample Size |
|---|---|
| -65° to -70°F | 10 |
| 65° to 75°F | 10 |
| 160° to 165°F | 10 |

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings C8845190 and C8845191.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Delay time.-Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1377-A483



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# SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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